

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMICO DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 05-11-2024	 CONSORZIO LaMMA
--	---	---

FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
October 2024



Contents

1	NIGHT1: Contingency tables	2
2	DAY1: Contingency tables	8
3	NIGHT2: Contingency tables	11
4	DAY2: Contingency tables	16
5	NIGHT3: Contingency tables	18
6	DAY3: Contingency tables	23

1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed		Observations		
ms^{-1}		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1594	759	111
	4.2 < ws < 7.6	1071	1314	638
	ws > 7.6	157	563	3633
Number of points 9840 ; PC=66.5%; EBD=2.7%; POD1=56.5%; POD2=49.8%; POD3=82.9%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed		Observations		
ms^{-1}		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2614	159	0
	4.2 < ws < 7.6	208	2336	190
	ws > 7.6	0	141	4192
Number of points 9840 ; PC=92.9%; EBD=0.0%; POD1=92.6%; POD2=88.6%; POD3=95.7%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed	Observations			
ms^{-1}	ws < 12.0	12.0 < ws < 18.0	ws > 18.0	
Model data	ws < 12.0	8029	124	0
	12.0 < ws < 18.0	93	1260	39
	ws > 18.0	0	61	234
Number of points 9840 ; PC=96.8%; EBD=0.0%; POD1=98.9%; POD2=87.2%; POD3=85.7%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	5427	247	0		92
	East	254	426	45		0
	South	34	119	838		83
	West	248	46	7		100
Number of points 7966 ; PC=85.2%; EBD=1.0%; POD1=91.0%; POD2=50.8%; POD3=94.2%; POD4=36.4%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	5893	55	0		20
	East	52	756	26		0
	South	0	27	859		5
	West	18	0	5		250
Number of points 7966 ; PC=97.4%; EBD=0.0%; POD1=98.8%; POD2=90.2%; POD3=96.5%; POD4=90.9%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3370	1019	9
	7.1 < rh < 15.3	1492	1883	224
	rh > 15.3	128	363	1081
Number of points 9569 ; PC=66.2%; EBD=1.4%; POD1=67.5%; POD2=57.7%; POD3=82.3%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	4764	187	0
	7.1 < rh < 15.3	226	3040	32
	rh > 15.3	0	38	1282
Number of points 9569 ; PC=95.0%; EBD=0.0%; POD1=95.5%; POD2=93.1%; POD3=97.6%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	9527	6	0
	50.0 < rh < 70.0	3	33	0
	rh > 70.0	0	0	0
Number of points 9569 ; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=84.6%; POD3=NaN%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	5410	951	10
	1.9 < pwv < 3.5	344	1713	180
	pwv > 3.5	0	65	1063
Number of points 9736 ; PC=84.1%; EBD=0.1%; POD1=94.0%; POD2=62.8%; POD3=84.8%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	5680	95	0
	1.9 < pwv < 3.5	74	2616	17
	pwv > 3.5	0	18	1236
Number of points 9736 ; PC=97.9%; EBD=0.0%; POD1=98.7%; POD2=95.9%; POD3=98.6%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	760	694	240
	0.6 < see < 0.8	520	672	724
	see > 0.8	59	270	756
Number of points 4695 ; PC=46.6%; EBD=6.4%; POD1=56.8%; POD2=41.1%; POD3=44.0%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1333	258	4
	0.6 < see < 0.8	157	1286	133
	see > 0.8	2	165	1737
Number of points 4695 ; PC=85.8%; EBD=0.1%; POD1=89.3%; POD2=75.2%; POD3=92.7%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1287	0	162
	0.6 < see < 0.8	0	1583	162
	see > 0.8	52	53	1558
Number of points 4695 ; PC=94.3%; EBD=4.6%; POD1=96.1%; POD2=96.8%; POD3=82.8%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1490	3	4
	0.6 < see < 0.8	0	1701	4
	see > 0.8	2	5	1870
Number of points 4695 ; PC=99.7%; EBD=0.1%; POD1=99.9%; POD2=99.5%; POD3=99.6%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1058	632	45
	$3.4 < \tau < 5.8$	334	976	352
	$\tau > 5.8$	19	238	479
Number of points 4133 ; PC=60.8%; EBD=1.5%; POD1=75.0%; POD2=52.9%; POD3=54.7%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1331	136	1
	$3.4 < \tau < 5.8$	80	1613	111
	$\tau > 5.8$	0	97	764
Number of points 4133 ; PC=89.7%; EBD=0.0%; POD1=94.3%; POD2=87.4%; POD3=87.2%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1320	135	170
	$3.4 < \tau < 5.8$	72	1632	170
	$\tau > 5.8$	19	79	706
Number of points 4133 ; PC=88.5%; EBD=4.6%; POD1=93.6%; POD2=88.4%; POD3=67.5%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1411	0	11
	$3.4 < \tau < 5.8$	0	1840	11
	$\tau > 5.8$	0	6	865
Number of points 4133 ; PC=99.6%; EBD=0.3%; POD1=100.0%; POD2=99.7%; POD3=97.5%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	650	700
	0.6 < glf < 0.7	452	656
	glf > 0.7	76	312
Number of points 4259 ; PC=39.4%; EBD=9.6%; POD1=55.2%; POD2=39.3%; POD3=26.4%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	999	226
	0.6 < glf < 0.7	172	1234
	glf > 0.7	7	208
Number of points 4259 ; PC=80.1%; EBD=0.3%; POD1=84.8%; POD2=74.0%; POD3=83.4%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1091	117
	0.6 < glf < 0.7	11	1533
	glf > 0.7	76	18
Number of points 4259 ; PC=86.0%; EBD=10.6%; POD1=92.6%; POD2=91.9%; POD3=58.9%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1168	14
	0.6 < glf < 0.7	3	1653
	glf > 0.7	7	1
Number of points 4259 ; PC=99.3%; EBD=0.3%; POD1=99.2%; POD2=99.1%; POD3=99.3%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	1889	552	108
	4.6 < ws < 7.4	1309	1646	598
	ws > 7.4	122	509	3506
Number of points 10239 ; PC=68.8%; EBD=2.2%; POD1=56.9%; POD2=60.8%; POD3=83.2%				

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed		Observations		
ms^{-1}		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2960	217	3
	4.6 < ws < 7.4	358	2315	143
	ws > 7.4	2	175	4066
Number of points 10239 ; PC=91.2%; EBD=0.0%; POD1=89.2%; POD2=85.5%; POD3=96.5%				

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations			
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0	
	ws < 12.0	7909	151	0
Model data	12.0 < ws < 18.0	106	1519	67
	ws > 18.0	0	64	423
Number of points 10239 ; PC=96.2%; EBD=0.0%; POD1=98.7%; POD2=87.6%; POD3=86.3%				

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	4742	303	0		110
	East	53	323	46		0
	South	25	17	564		238
	West	1061	18	70		1377
Number of points 8947 ; PC=78.3%; EBD=0.5%; POD1=80.6%; POD2=48.9%; POD3=82.9%; POD4=79.8%						

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	5723	45	2		99
	East	44	609	8		0
	South	0	5	651		18
	West	114	2	19		1608
Number of points 8947 ; PC=96.0%; EBD=0.0%; POD1=97.3%; POD2=92.1%; POD3=95.7%; POD4=93.2%						

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4264	990	8
	8.9 < rh < 17.9	1363	2108	301
	rh > 17.9	34	207	914
Number of points 10189 ; PC=71.5%; EBD=0.4%; POD1=75.3%; POD2=63.8%; POD3=74.7%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	5472	183	0
	8.9 < rh < 17.9	189	3081	33
	rh > 17.9	0	41	1190
Number of points 10189 ; PC=95.6%; EBD=0.0%; POD1=96.7%; POD2=93.2%; POD3=97.3%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	10109	10	0
	50.0 < rh < 70.0	7	59	4
	rh > 70.0	0	0	0
Number of points 10189 ; PC=99.8%; EBD=0.0%; POD1=99.9%; POD2=85.5%; POD3=0.0%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5868	612	0
	2.1 < pwv < 3.7	393	1795	344
	pwv > 3.7	0	86	1060
Number of points 10158 ; PC=85.9%; EBD=0.0%; POD1=93.7%; POD2=72.0%; POD3=75.5%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6180	49	0
	2.1 < pwv < 3.7	81	2424	23
	pwv > 3.7	0	20	1381
Number of points 10158 ; PC=98.3%; EBD=0.0%; POD1=98.7%; POD2=97.2%; POD3=98.4%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1759	816	101
	4.2 < ws < 7.6	1038	1290	663
	ws > 7.6	232	642	3643
Number of points 10184 ; PC=65.7%; EBD=3.3%; POD1=58.1%; POD2=46.9%; POD3=82.7%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5548	265	16
	East	239	453	90
	South	38	109	819
	West	226	40	13

Number of points 8166 ; PC=85.4%; EBD=1.2%; POD1=91.7%; POD2=52.2%; POD3=87.3%; POD4=49.0%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3443	986
	7.1 < rh < 15.3	1667	2010
	rh > 15.3	115	368

Number of points 9907 ; PC=66.3%; EBD=1.2%; POD1=65.9%; POD2=59.8%; POD3=85.0%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	5489	999
	1.9 < pwv < 3.5	426	1851
	pwv > 3.5	0	56

Number of points 10074 ; PC=81.8%; EBD=0.4%; POD1=92.8%; POD2=63.7%; POD3=71.9%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	972	871	342
	0.6 < see < 0.8	755	1036	942
	see > 0.8	113	411	1463
Number of points 6905 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1753	0	224
	0.6 < see < 0.8	0	2202	224
	see > 0.8	87	116	2523
Number of points 6905 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1944	941	91
	$3.4 < \tau < 5.8$	406	1201	519
	$\tau > 5.8$	39	399	640
Number of points 6180 ; PC=61.2%; EBD=2.1%; POD1=81.4%; POD2=47.3%; POD3=51.2%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2256	200	313
	$3.4 < \tau < 5.8$	94	2203	313
	$\tau > 5.8$	39	138	937
Number of points 6180 ; PC=87.3%; EBD=5.7%; POD1=94.4%; POD2=86.7%; POD3=59.9%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	854	957
	0.6 < glf < 0.7	859	1213
	glf > 0.7	129	340
Number of points 6369 ; PC=41.8%; EBD=9.9%; POD1=46.4%; POD2=48.3%; POD3=29.4%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1681	226
	0.6 < glf < 0.7	32	2248
	glf > 0.7	129	36
Number of points 6369 ; PC=84.5%; EBD=10.9%; POD1=91.3%; POD2=89.6%; POD3=57.4%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	1898	550	130
	4.6 < ws < 7.4	1313	1701	569
	ws > 7.4	121	456	3513
Number of points 10251 ; PC=69.4%; EBD=2.4%; POD1=57.0%; POD2=62.8%; POD3=83.4%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4772	366	13
	East	61	247	34
	South	36	41	552
	West	1012	7	80

Number of points 8947 ; PC=77.7%; EBD=0.6%; POD1=81.1%; POD2=37.4%; POD3=81.3%; POD4=80.2%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3992	871
	8.9 < rh < 17.9	1633	2184
	rh > 17.9	36	250

Number of points 10189 ; PC=69.1%; EBD=0.5%; POD1=70.5%; POD2=66.1%; POD3=70.3%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5696	658
	2.1 < pwv < 3.7	557	1750
	pwv > 3.7	8	85

Number of points 10158 ; PC=83.2%; EBD=0.2%; POD1=91.0%; POD2=70.2%; POD3=71.4%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1386	853	109
	4.2 < ws < 7.6	1285	1159	685
	ws > 7.6	280	756	3604
Number of points 10117 ; PC=60.8%; EBD=3.8%; POD1=47.0%; POD2=41.9%; POD3=81.9%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5440	267	16
	East	372	424	111
	South	115	163	759
	West	165	20	9

Number of points 8171 ; PC=82.1%; EBD=1.8%; POD1=89.3%; POD2=48.5%; POD3=84.8%; POD4=27.7%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3243	958
	7.1 < rh < 15.3	1771	2058
	rh > 15.3	173	319

Number of points 9840 ; PC=64.7%; EBD=2.0%; POD1=62.5%; POD2=61.7%; POD3=80.8%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	5456	1106
	1.9 < pwv < 3.5	359	1623
	pwv > 3.5	0	159

Number of points 9956 ; PC=78.6%; EBD=0.4%; POD1=93.8%; POD2=56.2%; POD3=59.7%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	945	705	309
	0.6 < see < 0.8	720	1039	892
	see > 0.8	191	560	1492
Number of points 6853 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1709	0	186
	0.6 < see < 0.8	0	2194	186
	see > 0.8	147	110	2507
Number of points 6853 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1871	913	42
	$3.4 < \tau < 5.8$	487	1266	600
	$\tau > 5.8$	28	331	593
Number of points 6131 ; PC=60.8%; EBD=1.1%; POD1=78.4%; POD2=50.4%; POD3=48.0%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2201	168	303
	$3.4 < \tau < 5.8$	157	2191	303
	$\tau > 5.8$	28	151	932
Number of points 6131 ; PC=86.8%; EBD=5.4%; POD1=92.2%; POD2=87.3%; POD3=60.6%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	820	859	415
	0.6 < glf < 0.7	873	1276	916
	glf > 0.7	139	358	664
Number of points 6320 ; PC=43.7%; EBD=8.8%; POD1=44.8%; POD2=51.2%; POD3=33.3%				

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1654	171	459
	0.6 < glf < 0.7	39	2292	400
	glf > 0.7	139	30	1536
Number of points 6320 ; PC=86.7%; EBD=9.5%; POD1=90.3%; POD2=91.9%; POD3=64.1%				

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	1740	503	122
	4.6 < ws < 7.4	1431	1652	560
	ws > 7.4	161	552	3530
Number of points 10251 ; PC=67.5%; EBD=2.8%; POD1=52.2%; POD2=61.0%; POD3=83.8%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4644	368	20
	East	56	244	32
	South	87	34	536
	West	1094	15	91

Number of points 8947 ; PC=74.9%; EBD=1.4%; POD1=79.0%; POD2=36.9%; POD3=78.9%; POD4=74.0%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3749	750
	8.9 < rh < 17.9	1815	2240
	rh > 17.9	97	315

Number of points 10189 ; PC=67.2%; EBD=1.0%; POD1=66.2%; POD2=67.8%; POD3=69.8%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5605	776
	2.1 < pwv < 3.7	648	1593
	pwv > 3.7	8	124

Number of points 10158 ; PC=81.0%; EBD=0.1%; POD1=89.5%; POD2=63.9%; POD3=73.6%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration